

Computer Vision Systems – Final Revision

Part 1: True / False Questions

Sheet 1

1. Pixel processing operates on groups of pixels simultaneously. **False**
2. Smoothing suppresses high-frequency components. **True**
3. Gradient operators suppress high frequencies. **False**
4. Averaging filters are nonlinear filters. **False**
5. Mean filtering is implemented using convolution. **True**
6. Averaging filters preserve edges perfectly. **False**
7. Gaussian filters are linear smoothing filters. **True**
8. Nonlinear filters cannot be expressed using convolution. **True**
9. Median filters are effective against salt-and-pepper noise. **True**
10. Median filtering requires sorting pixel values. **True**
11. Histogram shows spatial location of pixels. **False**
12. Histogram shows intensity distribution. **True**
13. Histogram is global information of an image. **True**
14. Histogram equalization reduces contrast. **False**
15. Linear filters use weighted sums of pixels. **True**
16. Image smoothing may blur important edges. **True**
17. Smoothing operators are used to suppress low frequencies and enhance edges. **False**
18. A serious disadvantage of the averaging (mean) filter is the blurring of edges. **True**
19. The Gaussian filter is a type of non-linear filter. **False**
20. The median filter replaces a pixel's value with the average of its neighbors. **False**
21. The median filter is very effective at removing Gaussian noise. **False**
22. A histogram provides local information about a specific pixel's neighborhood. **False**
23. Gradient operators are used for image smoothing. **False**
24. One of the disadvantages of the median filter is that it can damage sharp corners. **True**
25. Histogram Equalization is a linear filtering technique. **False**
26. The box filter is an example of a non-linear filter. **False**

Sheet 2

1. The gradient operator's magnitude gives edge orientation. **False**
2. The Roberts operator is larger than the Sobel operator and therefore less sensitive to noise.
False
3. Larger derivative operators incorporate implicit smoothing and have better noise resilience.
True
4. Hysteresis thresholding uses only a single threshold to classify edge pixels. **False**
5. The Laplacian operator provides both edge magnitude and orientation. **False**
6. Noise is problematic for edge detection because it also contains rapid changes. **True**
7. Convolving a signal with a Gaussian then taking the derivative is equivalent to convolving with the derivative of the Gaussian. **True**
8. The Canny edge detector applies a 2D Laplacian over the entire image to find edges. **False**
9. Increasing σ in the Canny detector results in detecting more fine-grained edges. **False**
10. In the gradient space (I_x, I_y) , a flat region produces a cluster that is long and thin. **False**
11. In the gradient space, an edge produces one large and one small eigenvalue λ . **True**
12. A corner produces two small eigenvalues in the gradient space. **False**
13. The Harris response R is designed such that corners yield large positive values. **True**
14. Non-maximal suppression is used only for corner detection and not for other vision tasks.
False
15. The Laplacian operator requires two separate convolutions followed by a nonlinear operation. **False**
16. The gradient operator is more computationally efficient than the Laplacian because it uses a smaller kernel. **False**

Sheet 3

1. Active contours can only be used for static images, not video sequences. **False**
2. In active contours, the image term alone is sufficient to guarantee a smooth, noise-free boundary. **False**

3. The internal bending energy of a contour consists of elastic and smoothness terms. **True**
4. The elastic term in active contours is based on the second derivative of the contour. **False**
5. The smoothness term in active contours is based on the curvature (second derivative). **True**
6. A large β (smoothness weight) makes the contour more sensitive to local edge details. **False**
Active contours are widely used in medical imaging. **True**
7. The Hough transform requires edges to be perfectly connected to work. **False**
8. In the (m, c) Hough transform, a line in image space maps to a point in parameter space. **True**
9. The (m, c) parameterization can represent vertical lines without any issue. **False**
10. In the (ρ, θ) parameterization of straight lines, ρ is unbounded. **False**
11. In circle detection with known radius, a point maps to a circle in parameter space. **True**
12. Using edge direction information in circle detection reduces the number of votes. **True**
13. For circles with unknown radius, the parameter space is 2D. **False**
14. The generalized Hough transform can detect arbitrary shapes without a prior model. **False**
15. In the generalized Hough transform, the φ -table index is the edge direction φ . **True**
16. The generalized Hough transform can handle occlusion and missing boundary data. **True**

Sheet 4

1. SIFT can be used for image stitching and object recognition. **True**
2. The SIFT detector uses the Laplacian of Gaussian (LoG) directly without any normalization.
False
3. The characteristic scale of a blob is proportional to its width. **True**
4. Scale space is created by convolving an image with Gaussians of different σ values. **True**
5. The Difference of Gaussian (DoG) is an exact implementation of the NLoG. **False**
6. The DoG can be computed by subtracting consecutive images in the scale space stack. **True**
7. In SIFT, all detected extrema are kept regardless of their strength. **False**
8. The circles overlaid on SIFT results have radii proportional to the characteristic scale of the feature. **True**
9. To achieve scale invariance, SIFT uses the characteristic scale to normalize the image patch.
True
10. For rotation invariance, SIFT uses the gradient magnitude histogram. **False**

11. The principal orientation corresponds to the gradient direction with the maximum histogram value. **True**
12. The SIFT descriptor is constructed from raw pixel intensities. **False**
13. The SIFT descriptor is computed on a scale- and rotation-normalized window. **True**
14. When comparing two SIFT descriptors, L2 distance gives a perfect match when the value is 1. **False**
15. When comparing two SIFT descriptors, normalized correlation yields 1 for a perfect match. **True**
16. When comparing two SIFT descriptors, the intersection metric gives a larger value for a better match. **True**
17. SIFT is highly effective for recognizing 3D objects from arbitrary viewpoints. **False**
18. SIFT can handle occlusion and clutter to some extent. **True**
19. The NLoG operator in 2D produces a stack of images indexed by (x, y, σ) . **True**
20. The SIFT descriptor is sensitive to illumination changes. **False**
21. SIFT can be used to create panoramas by matching features between overlapping images. **True**

Part 2: Multiple Choice Questions (MCQ)

Sheet 1

1. **Convolution in image processing is used mainly for:**

- A) Filtering
- B) Compressing
- C) Storing
- D) Encrypting

ANSWER: A

2. **Image smoothing techniques aim to:**

- A) Increase noise
- B) Reduce noise
- C) Increase brightness
- D) Increase resolution

ANSWER: B

3. **Smoothing suppresses:**

- A) High frequencies
- B) Low frequencies
- C) Colors
- D) Pixel coordinates

ANSWER: A

4. **Gradient operators mainly preserve:**

- A) Low frequencies
- B) High frequencies (edges)
- C) Color channels
- D) Pixel locations

ANSWER: B

5. **Averaging filters belong to:**

- A) Linear filters
- B) Nonlinear filters
- C) Frequency filters
- D) Histogram filters

ANSWER: A

6. **Mean filters are implemented using:**

- A) Sorting
- B) Convolution
- C) Fourier transform
- D) Histogram equalization

ANSWER: B

7. **The main disadvantage of averaging filters is:**

- A) They increase noise
- B) They blur edges
- C) They reduce brightness
- D) They increase resolution

ANSWER: B

8. **Gaussian filters are used mainly for:**

- A) Edge detection
- B) Noise reduction
- C) Color conversion
- D) Object detection

ANSWER: B

9. **Nonlinear filters cannot be implemented using:**

- A) Sorting
- B) Convolution
- C) Algorithms
- D) Neighborhood operations

ANSWER: B

10. Median filtering is particularly effective for:

- A) Gaussian noise
- B) Salt and pepper noise
- C) Periodic noise
- D) Random noise

ANSWER: B

11. One disadvantage of the median filter is:

- A) It increases noise
- B) It damages thin lines and fine details
- C) It increases brightness
- D) It compresses images

ANSWER: B

12. Histogram represents:

- A) Pixel positions
- B) Frequency distribution of intensity values
- C) Image resolution
- D) Image coordinates

ANSWER: B

13. Histogram provides:

- A) Global image information
- B) Local pixel values
- C) Pixel coordinates
- D) Filter values

ANSWER: A

14. Histogram equalization is used to:

- A) Reduce noise
- B) Improve contrast
- C) Reduce image size
- D) Detect edges

ANSWER: B

15. What is the primary purpose of image smoothing techniques?

- A) To enhance edges
- B) To suppress high frequencies (noise)
- C) To increase image contrast
- D) To amplify low frequencies

ANSWER: B

16. What is the main disadvantage of using averaging (mean) filters for smoothing?

- A) They are computationally very slow
- B) They cannot remove any type of noise
- C) They blur sharp edges
- D) They introduce "salt and pepper" noise

ANSWER: C

17. Which type of filter is very effective for removing "salt and pepper" noise?

- A) Mean Filter
- B) Gaussian Filter
- C) Median Filter
- D) Box Filter

ANSWER: C

18. Gradient operators are designed to suppress which part of the frequency domain?

- A) High frequencies
- B) Low frequencies
- C) Mid frequencies
- D) All frequencies equally

ANSWER: B

19. What information does an image histogram provide?

- A) The location of edges in the image
- B) The frequency of each brightness (intensity) value in the image
- C) The result of a convolution operation
- D) The noise level of the image

ANSWER: B

20. Non-linear filters, such as the median filter, are characterized by the fact that they:

- A) Can be implemented as a convolution
- B) Are faster than linear filters
- C) Are more algorithmic and cannot be implemented as convolutions
- D) Always preserve fine details perfectly

ANSWER: C

21. Besides blurring edges, what is another disadvantage of the median filter?

- A) It is ineffective against random noise
- B) It can erase fine details like thin lines
- C) It always creates new, artificial pixel values
- D) It enhances "salt and pepper" noise

ANSWER: B

22. The process of Histogram Equalization primarily aims to:

- A) Blur the image
- B) Detect edges in the image
- C) Redistribute the intensity distribution to enhance contrast
- D) Shift the image's pixel values

ANSWER: C

23. Which filter is a type of linear smoothing filter?

- A) Median Filter
- B) Gaussian Filter
- C) Min Filter
- D) Max Filter

ANSWER: B

24. The Gaussian filter is a specific example of which broader category of filters?

- A) Non-linear filters
- B) Gradient operators
- C) Averaging (Linear) filters
- D) Histogram-based filters

ANSWER: C

25. What is a disadvantage of the median filter related to computation?

- A) It is not effective for any type of noise.
- B) Sorting values to find the median is time-consuming compared to averaging.
- C) It requires complex convolution operations.
- D) It cannot be implemented in software.

ANSWER: B

Sheet 2

1. **The magnitude of an edge using the gradient operator is computed as:**

- A) $|I_x| + |I_y|$
- B) $\max(I_x, I_y)$
- C) $\sqrt{I_x^2 + I_y^2}$
- D) $\tan^{-1}\left(\frac{I_y}{I_x}\right)$

ANSWER: C

2. **Which of the following gradient operators is the smallest and therefore most sensitive to noise?**

- A) Sobel (3x3)
- B) Prewitt
- C) Roberts
- D) Sobel (5x5)

ANSWER: C

3. **Larger gradient operators have better noise resilience but suffer from:**

- A) Higher computational cost only
- B) Poor localization
- C) Inability to detect vertical edges
- D) Loss of magnitude information

ANSWER: B

4. **In hysteresis-based edge thresholding, a pixel with magnitude between T_0 and T_1 is declared an edge if:**

- A) Its magnitude is above the mean
- B) It has a neighboring pixel that is definitely an edge
- C) Its orientation matches the global gradient
- D) It is a local maximum

ANSWER: B

5. **In the Laplacian operator, edges are detected at:**

- A) Local maxima of the Laplacian
- B) Local minima of the Laplacian
- C) Zero-crossings of the Laplacian
- D) Points where the Laplacian is constant

ANSWER: C

6. **The Laplacian operator provides:**

- A) Edge location, magnitude, and direction
- B) Edge location only
- C) Edge magnitude only
- D) Edge direction only

ANSWER: B

7. **To compute the Laplacian on a discrete image, the minimum window size required is:**

- A) 2x2
- B) 3x3
- C) 5x5
- D) 1x1

ANSWER: B

8. **The Canny edge detector combines smoothing, gradient computation, and:**

- A) Harris response
- B) 1D Laplacian along the gradient direction
- C) Median filtering
- D) Histogram equalization

ANSWER: B

9. **In the Canny detector, the parameter σ (Gaussian width) controls:**

- A) The orientation of edges
- B) The scale (resolution) of edges detected
- C) The threshold for hysteresis
- D) The size of the non-maximal suppression window

ANSWER: B

10. **In the gradient space (I_x, I_y) used for corner detection, a flat region produces:**

- A) A long, thin cluster
- B) Two separate clusters
- C) A compact cluster near the origin
- D) A circular cluster far from the origin

ANSWER: C

11. **In the gradient space, an edge region produces:**

- A) A compact cluster near the origin
- B) A long, thin cluster
- C) Two separate clusters
- D) A random scatter

ANSWER: B

12. **For a corner region in gradient space, both eigenvalues λ_1 and λ_2 are:**

- A) Small
- B) One large, one small
- C) Both large
- D) Zero

ANSWER: C

13. **The Harris corner detector uses a response function R to map:**

- A) I_x and I_y to a single number
- B) λ_1 and λ_2 to a single number
- C) Gradient magnitude and orientation to a score
- D) Laplacian and gradient to a score

ANSWER: B

14. **After computing the Harris response R, which technique is used to find exact corner locations?**

- A) Hysteresis thresholding
- B) Gaussian smoothing
- C) Non-maximal suppression
- D) Zero-crossing detection

ANSWER: C

15. **Non-maximal suppression works by:**

- A) Averaging values in a window
- B) Retaining only the maximum value in a local window
- C) Applying a second derivative
- D) Thresholding at the mean

ANSWER: B

16. **Which of the following is an advantage of the gradient operator over the Laplacian?**

- A) It is a single linear convolution
- B) It provides edge direction/orientation
- C) It does not require thresholding
- D) It is less sensitive to noise

ANSWER: B

17. **The Laplacian of Gaussian (LoG) operator combines:**

- A) Smoothing with a Gaussian and then applying the Laplacian
- B) Gradient and Harris response
- C) Sobel and Prewitt operators
- D) Thresholding and non-maximal suppression

ANSWER: A

Sheet 3

1. **In active contours (snakes), the initial contour is typically:**

- A) Automatically generated from edge detection
- B) A fixed shape like a circle
- C) Provided by the user (e.g., sketched)
- D) Randomly initialized

ANSWER: C

2. **To make distant edges attract the contour, the gradient image is:**

- A) Thresholded
- B) Blurred with a Gaussian
- C) Differentiated twice
- D) Binarized

ANSWER: B

3. The image energy term E_{image} in active contours is designed to:

- A) Minimize the sum of squared gradient magnitudes
- B) Maximize the sum of squared gradient magnitudes
- C) Minimize the contour length
- D) Maximize the curvature

ANSWER: B

4. The elastic term in the active contour energy function minimizes:

- A) The second derivative of the contour
- B) The first derivative (rate of change) of the contour
- C) The third derivative
- D) The contour's area

ANSWER: B

5. The smoothness term in active contours minimizes:

- A) The first derivative
- B) The second derivative (curvature)
- C) The distance to edges
- D) The contour's perimeter

ANSWER: B

6. A large value of α (elasticity weight) in active contours makes the contour:

- A) More sensitive to image edges
- B) Behave like a stiffer rubber band (contract more)
- C) Expand outward
- D) Ignore smoothness

ANSWER: B

7. **The Hough transform is particularly robust against which three problems?**

- A) Blur, low contrast, and high resolution
- B) Incomplete data, noise, and occlusion
- C) Motion blur, saturation, and compression
- D) Rotation, scaling, and translation

ANSWER: B

8. **In the Hough transform for lines using (m, c) parameterization, a point in image space maps to:**

- A) A point in parameter space
- B) A line in parameter space
- C) A sinusoid in parameter space
- D) A circle in parameter space

ANSWER: B

9. **The intersection of two lines in (m, c) parameter space corresponds to:**

- A) Two different lines in image space
- B) The line passing through the two corresponding image points
- C) A point in image space
- D) No valid interpretation

ANSWER: B

10. **Why is the (m, c) parameterization problematic for the Hough transform?**

- A) It cannot represent vertical lines (infinite slope)
- B) It requires too much memory
- C) It is nonlinear
- D) It only works for circles

ANSWER: A

11. In the (ρ, θ) parameterization, a point in image space maps to:

- A) A line
- B) A sinusoid
- C) A circle
- D) A point

ANSWER: B

12. In the Hough transform for circles of known radius, a point in image space maps to:

- A) A line
- B) A sinusoid
- C) A circle in parameter space
- D) A cone in parameter space

ANSWER: C

13. If edge direction is known for circle detection, voting reduces from a full circle to:

- A) One point
- B) Two points
- C) A line
- D) A sinusoid

ANSWER: B

14. For circles with unknown radius and unknown edge direction, the parameter space is:

- A) 2D
- B) 3D
- C) 4D
- D) 1D

ANSWER: B

15. The Generalized Hough Transform uses a model called:

- A) The accumulator array
- B) The φ -table
- C) The snake matrix
- D) The distance transform

ANSWER: B

16. In the Generalized Hough Transform, the reference point's location is found by:

- A) Minimizing an energy function
- B) Voting in a 2D accumulator array
- C) Fitting a polynomial
- D) Thresholding the edge map

ANSWER: B

17. To handle unknown scale and rotation in the Generalized Hough Transform, the accumulator array becomes:

- A) 2D
- B) 3D
- C) 4D
- D) 5D

ANSWER: C

Sheet 4

1. For a blob-like feature to be useful, we need to determine all of the following EXCEPT:

- a) Its location
- b) Its size
- c) Its color
- d) Its principal orientation

Answer: c

2. The characteristic scale of a blob is determined by:

- a) The peak intensity of the blob
- b) The σ value at which the NLoG response attains a maximum
- c) The width of the blob at half maximum
- d) The average gradient magnitude

Answer: b**3. The normalized Laplacian of Gaussian (NLoG) operator is given by:**

- a) $\sigma \nabla^2 n_\sigma * I(x, y)$
- b) $\sigma^2 \nabla^2 n_\sigma * I(x, y)$
- c) $\nabla^2 n_\sigma * I(x, y)$
- d) $\sigma^3 \nabla^2 n_\sigma * I(x, y)$

Answer: b**4. In the scale space representation $S(x, y, \sigma) = n(x, y, \sigma) * I(x, y)$, σ represents:**

- a) The image intensity
- b) The scale (width of Gaussian)
- c) The edge orientation
- d) The corner response

Answer: b**5. The discrete scales for creating scale space are typically chosen as:**

- a) $\sigma_k = \sigma_0 + k \cdot s$
- b) $\sigma_k = \sigma_0 \cdot s^k$
- c) $\sigma_k = \frac{\sigma_0}{s^k}$
- d) $\sigma_k = \sigma_0 + s \cdot k^2$

Answer: b

6. The Difference of Gaussian (DoG) operator approximates:

- a) The gradient of Gaussian
- b) The normalized Laplacian of Gaussian (NLoG)
- c) The Harris corner response
- d) The Sobel operator

Answer: b

7. In the SIFT detector, extrema are found in the DoG stack by:

- a) Thresholding each image independently
- b) Using a small $n \times n \times n$ window over the volume
- c) Applying a Fourier transform
- d) Computing the mean of each scale

Answer: b

8. To achieve rotation invariance, SIFT computes:

- a) The average intensity of the blob
- b) The principal orientation from gradient orientation histogram
- c) The centroid of the blob
- d) The second moment matrix

Answer: b

9. When computing the principal orientation, SIFT ignores gradient magnitude to:

- a) Save computation
- b) Reduce sensitivity to illumination and camera gain
- c) Increase scale invariance
- d) Improve localization

Answer: b

10. The SIFT descriptor is constructed from:

- a) Raw pixel intensities
- b) Gradient orientations in a normalized grid
- c) Edge magnitudes

d) Laplacian responses

Answer: b

11. The SIFT descriptor is a normalized histogram that is invariant to:

a) Scale, rotation, and illumination

b) Only rotation

c) Only scale

d) Only illumination

Answer: a

12. Which of the following metrics for comparing SIFT descriptors gives a perfect match when the value is 1?

a) L2 distance

b) Normalized correlation

c) Intersection

d) Manhattan distance

Answer: b

13. Which matching metric computes the sum of minimum values per bin?

a) L2 distance

b) Normalized correlation

c) Intersection

d) Euclidean distance

Answer: c

14. SIFT works best for recognizing:

a) Arbitrary 3D objects from any viewpoint

b) More or less planar (flat) objects

c) Only perfectly fronto-parallel objects

d) Only objects with uniform texture

Answer: b

15. When the viewpoint difference between two images of a 3D object increases to 90 degrees, SIFT matches:

- a) Increase significantly
- b) Remain constant
- c) Drop dramatically (hardly any matches)
- d) Become perfect

Answer: c

16. Which of the following is an application of SIFT?

- a) Face recognition
- b) Image stitching and panorama creation
- c) Speech recognition
- d) Audio processing

Answer: b

17. The NLoG operator is used for blob detection because it:

- a) Responds strongly at blob centers with characteristic scale
- b) Only responds to edges
- c) Is computationally cheaper than DoG
- d) Does not require convolution

Answer: a

18. The DoG approximation is efficient because it:

- a) Uses fewer convolutions
- b) Simply subtracts consecutive images in the scale space stack
- c) Requires no Gaussian smoothing
- d) Works only on binary images

Answer: b

Part 3: Essay Questions**Sheet 1****1. What is the main conceptual difference between Linear and Non-linear image filters?**

Give one example of each type.

ANSWER:

The main conceptual difference is in how they operate.

- **Linear filters** perform a linear operation, specifically a weighted sum of the pixels in a neighborhood, which is mathematically defined as convolution.

Example of a Linear Filter: Mean filter, Gaussian filter.

- **Non-linear filters**, on the other hand, are more algorithmic. Their output is not a simple weighted sum and they cannot be represented or implemented as a convolution. Their operation often involves sorting, logical decisions, or other non-linear rules.

Example of a Non-linear Filter: Median filter.

2. What is an image histogram? If an image is very dark, what would you expect its histogram to look like?**ANSWER:**

- An image histogram is a graph or plot that shows the frequency distribution of all the pixel intensity values in an image. The x-axis represents the possible intensity values (e.g., 0 to 255 for an 8-bit image), and the y-axis represents the number of pixels in the image that have that specific intensity value.
- If an image is very dark, most of its pixels will have low intensity values. Therefore, its histogram would be heavily skewed to the left side of the graph, with most of the "bars" or data points clustered near 0.

3. What does it mean that a histogram provides "global information" about an image?**ANSWER:**

It means that the histogram summarizes information from the entire image as a whole, rather than any specific part of it. It tells you, for example, that the image is generally dark or has high contrast, but it provides no information about *where* in the image the dark or bright pixels are located. Two completely different images (e.g., a picture of a cat and a picture of a landscape) could potentially have identical histograms.

4. What is the advantage of histogram equalization?**ANSWER:**

Histogram equalization improves image contrast by redistributing intensity values across the full intensity range.

5. Explain the effect of using a smoothing filter and using a gradient operator on an image.**ANSWER:**

- **Smoothing filters** aim to suppress high-frequency noise, which improves the image by making it less noisy. However, this comes at the cost of blurring sharp edges, which are also high-frequency information.
- **Gradient operators** do the opposite; they suppress low frequencies, which effectively highlights and sharpens edges.

Sheet 2**6. Compare between the gradient operator and the Laplacian operator for edge detection.****Answer:**

- The gradient operator provides edge location, magnitude, and orientation. It requires two convolutions (for I_x and I_y) followed by a nonlinear operation for magnitude and orientation.
- The Laplacian provides only edge location (via zero-crossings) and no orientation. It requires a single linear convolution.
- The gradient needs thresholding to declare edges; the Laplacian needs zero-crossing detection.

7. Describe the three performance criteria for a good edge detector. Why is there a trade-off between noise sensitivity and localization?

Answer:

- The three criteria are:
 - (1) High detection rate.
 - (2) Good localization.
 - (3) Low noise sensitivity.
- The trade-off is that larger smoothing kernels suppress noise better (low noise sensitivity) but blur the edge, causing poor localization. Smaller kernels give excellent localization but are very sensitive to noise.

8. Why does the Canny edge detector uses the "best attributes" of both the gradient and Laplacian operators?

ANSWER:

Canny uses the gradient operator to compute magnitude and orientation (a strength of the gradient). Then it applies a 1D Laplacian (second derivative) along the gradient direction to find zero-crossings (a strength of the Laplacian for precise localization). This combination yields reliable edge detection with good localization.

9. Explain the concept of "scale space" in the context of the Canny edge detector. How does changing σ affect the output?

Answer:

Scale space refers to the idea that edges exist at different levels of detail or resolution. A small σ preserves fine details (e.g., pores in a brick), while a large σ suppresses fine details and only preserves coarse structures (e.g., brick borders). By increasing σ , the Canny detector produces fewer, more prominent edges corresponding to a coarser scale.

10. Explain why noise is particularly problematic for edge detection, and describe the standard solution for this problem.

Answer:

- Noise is problematic because it consists of rapid, random intensity changes, which are exactly what edge detectors (derivatives) are designed to respond to. This creates many false edges.
- The standard solution is to first smooth the image with a Gaussian filter to suppress noise before applying the derivative operator. Alternatively, one can convolve with the derivative of a Gaussian directly.

11. What is non-maximal suppression and why is it necessary after computing the Harris corner response?

Answer:

- Non-maximal suppression is a technique where a window slides over the response image, and only pixels that are the maximum value within that window are retained; all others are suppressed or set to zero.
- It is necessary because the Harris response produces large values not only at the exact corner location but also in its vicinity. Non-maximal suppression finds the local peaks, giving the precise corner locations.

Sheet 3

12. What are the three energy terms in an active contour (snake) formulation? Explain the physical meaning of each.

ANSWER:

(1) E_{image} : attracts contour to strong edges (gradient maxima).

(2) $E_{elastic}$: based on first derivative, makes the contour behave like a rubber band (contracts).

(3) E_{smooth} : based on second derivative (curvature), penalizes sharp bends, making the contour smooth like a metal strip.

$$E_{total} = E_{image} + \alpha E_{elastic} + \beta E_{smooth}$$

13. Explain the duality between image space and parameter space in the Hough transform for line detection using the (m, c) parameterization.

ANSWER:

- A point (x_i, y_i) in image space maps to a line $c = -m x_i + y_i$ in (m, c) parameter space.
- A line in image space (with fixed m, c) maps to a point in parameter space.
- This duality allows voting: each image point votes for all possible lines through it, and intersections (peaks) in parameter space correspond to lines in image space.

14. Why is the (ρ, θ) parameterization preferred over (m, c) for the Hough transform for line detection? What are the ranges of ρ and θ ?

ANSWER:

- (m, c) cannot represent vertical lines ($m = \infty$) and requires unbounded accumulator arrays.
- (ρ, θ) has finite ranges: $0 \leq \theta < \pi$ and ρ bounded by the image diagonal. It represents all lines uniformly, including vertical ones.

15. Describe the voting process in the Hough transform for detecting circles of known radius. What does a peak in the accumulator represent?

ANSWER:

- For each edge point (x_i, y_i) , vote for all possible centers (a, b) such that $(x_i - a)^2 + (y_i - b)^2 = r^2$, i.e., a circle in parameter space.
- After voting, a peak in the 2D accumulator array (a, b) indicates the center of a circle of radius r in the image that passes through many edge points.

16. How does knowing the edge direction ϕ reduce computational cost in circle detection?**ANSWER:**

- With edge direction ϕ , the center must lie along the line perpendicular to the edge at distance r in either direction.
- This gives only two possible centers per edge point instead of an entire circle of centers.
- So, in parameter space, we no longer need to vote along an entire circle, we only have to vote at two points on either side of the edge. Thus, we only need to increment two points in the accumulator array, which is computationally a major saving.
- The trade-off is that more work is done in image space (computing accurate edge directions) to save work in parameter space.

17. Explain the structure and purpose of the ϕ -table in the Generalized Hough Transform.

How is it created, and how is it used during detection?

ANSWER:

- The ϕ -table is an off-line model of the target shape.
- For each edge direction ϕ , the table stores a list of vectors $\vec{r}$ from boundary points with that direction to a chosen reference point (x_c, y_c) .
- During detection, for each image edge point with direction ϕ , the corresponding vectors are retrieved, and votes are cast for possible reference point locations.
- Peaks in the accumulator indicate detected objects.

18. What happens to the Generalized Hough Transform when unknown scale and rotation are added? Why does this become impractical?**ANSWER:**

- The accumulator array becomes 4D $(x_c, y_c, \text{scale}, \text{rotation})$. For each edge point, voting must be done over multiple scale and rotation combinations.
- As a result, memory and computation grows proportionally. This is rarely practical for real-world applications due to memory and computational constraints.

19. Compare between the Hough transform and active contours (snakes) for boundary detection. What types of problems is each best suited for?

ANSWER:

Hough transform:

- Best for detecting parametric shapes (lines, circles) in cluttered images.
- Works on disconnected edges, robust to noise and occlusion.
- Does not require initialization.
- Cannot easily handle arbitrary non-parametric shapes without generalization (which becomes expensive).

Active contours:

- Best for finding arbitrary, smooth boundaries when a rough initialization is available (e.g., user input or previous frame).
- Requires good initialization and can get stuck in local minima.
- Excellent for medical imaging and tracking deformable objects over time.

Sheet 4

20. Explain the concept of "characteristic scale" in blob detection. How is it determined using the NLoG operator?

Answer:

- The characteristic scale of a blob is the scale (σ value) at which the σ -normalized Laplacian of Gaussian (NLoG) response attains a local maximum at the blob's center.
- As σ increases, the response for a blob rises, peaks, then falls. The σ at the peak is proportional to the blob's width and becomes its characteristic scale.

21. Describe the Difference of Gaussian (DoG) approximation. Why is it used in SIFT instead of directly computing the NLoG?

Answer:

- DoG approximates the NLoG (up to a scale factor). It is computed by subtracting consecutive images in the scale space stack: $DoG = S(x, y, \sigma_k) - S(x, y, \sigma_{k-1})$.
- It is used because it is computationally efficient—it requires only image subtractions rather than full Laplacian convolutions at each scale.

22. How does SIFT achieve scale invariance? Explain the steps involved.

Answer:

SIFT achieves scale invariance by:

- (1) Detecting features at multiple scales using the DoG stack.
 - (2) Assigning each feature a characteristic scale (the σ at which it was detected).
 - (3) Normalizing the image patch around the feature to that scale before computing the descriptor.
- This makes the descriptor independent of the object's magnification.

23. How does SIFT achieve rotation invariance? Describe the process of finding the principal orientation.

Answer:

- For each feature, SIFT computes gradient orientations of all pixels in a window around the feature (scaled to its characteristic scale).
- A histogram of gradient orientations is created; the peak (or dominant peaks) becomes the principal orientation.
- The patch is then rotated so that this principal orientation points "north," making the descriptor rotation-invariant.

24. Compare and contrast two metrics for matching SIFT descriptors.**Answer:**

- L2 distance: computes $\sqrt{\sum (H_1(k) - H_2(k))^2}$. Smaller = better match. Perfect match = 0.
- Intersection: $\sum \min(H_1(k), H_2(k))$. Larger = better match. Measures overlap between histograms.

where $H_1(k)$ and $H_2(k)$ are two arrays of data of length N .

25. Name at least three applications of SIFT and briefly explain how SIFT is used in each.**Answer:**

- **Object recognition:** SIFT features from a reference object are matched to features in a scene to detect the object despite scale, rotation, clutter, and occlusion.
- **Image stitching / panorama creation:** SIFT features are detected in overlapping images, matched, and used to geometrically warp images into alignment.
- **Scene collages:** SIFT matches between images allow them to be overlaid (translated, scaled, rotated) to create a composite visualization.